VANCOMYCIN

Infuse slowly — flushing syndrome

Monitor trough levels

CHEMICAL NAME Glycopeptide antibiotic ($C_{66}H_{75}Cl_2N_9O_{24}$) from *Amycolatopsis orientalis*, as hydrochloride

CLASS Glycopeptide antibiotic

TRADE NAMES Vancocin · Vanco · Vancomycin HCl injection

THERAPEUTIC USES / INDICATIONS

- MRSA infections: bacteraemia, pneumonia, skin, bone and joint infections.
- Serious Gram-positive infections in penicillin allergy.
- Endocarditis (with gentamicin/rifampicin per guideline); meningitis (with others).
- Oral: *C. difficile* colitis (not absorbed).

ADMINISTRATION

- **IV infusion:** max 10 mg/min (at least 60 min per 1 g); dilute adequately.
- **PO** capsules or oral solution for *C. difficile* only.
- Irritant — use large vein; avoid extravasation.
- Take trough level before the 4th dose (or AUC monitoring per protocol).

DOSAGES

- **Adult IV:** 15–20 mg/kg every 8–12 h (loading 25–30 mg/kg in severe infection); target trough 10–20 mg/L (15–20 in severe).
- **C. difficile (PO):** 125 mg four times daily for 10 days.
- **Children IV:** 10–15 mg/kg every 6 h.
- Adjust by levels and renal function.

PHARMACOKINETICS

- **Onset:** IV immediate. Not absorbed orally (acts locally in gut).
- Not metabolised; poor CSF penetration unless inflamed.
- Excreted unchanged in urine. **t½:** 4–6 h (up to 7 days in renal failure).

PHARMACODYNAMICS

- Binds **D-alanyl-D-alanine** of cell-wall precursors → blocks peptidoglycan synthesis → bactericidal (slowly).
- Active only against Gram-positive bacteria, incl. MRSA, *C. difficile*, enterococci (not VRE).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Vancomycin flushing ("red man") syndrome: flushing, itching, rash of face/neck/upper body, hypotension — from rapid infusion (histamine release).
- Phlebitis.
- Nausea; bitter taste (oral).

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (↑ with high troughs, piperacillin–tazobactam, aminoglycosides).
- Ototoxicity (rare).
- Neutropenia, thrombocytopenia (prolonged).
- DRESS, SJS; anaphylaxis.

CONTRAINDICATIONS & CAUTIONS

- Vancomycin hypersensitivity.
- **Caution:** renal impairment, hearing loss, concurrent nephrotoxic drugs, elderly.

KEY INTERACTIONS

- **Aminoglycosides, piperacillin–tazobactam, loop diuretics, amphotericin, ciclosporin, NSAIDs:** ↑ nephro/ototoxicity.
- **Neuromuscular blockers:** enhanced block.

NURSING MANAGEMENT

- Use infusion pump; never infuse faster than 10 mg/min.
- If flushing occurs: stop/slow infusion, give antihistamine as prescribed, restart slower.
- Time trough levels correctly; check results before next dose.
- Monitor creatinine, urine output, CBC.
- Monitor IV site; ask about hearing.

PATIENT EDUCATION

- Report flushing, itching or dizziness during the drip.
- Report reduced urine, ringing in ears, rash.
- Oral vancomycin works only in the gut — take all doses.

KEY POINTS

- Glycopeptide for MRSA and serious Gram-positives.
- Monitor levels and kidney function.
- Infuse slowly; flushing syndrome.
- Oral form only for C. difficile.